SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

# Technology Stack

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| **Date** | 7 July 2026 |
| **Team ID** | xxxxxx |
| **Project Name** | EduGenie Google Gemini Powered Learning Assistant |
| **Maximum Marks** | 2 Marks |

**Define Your Technology Stack:**

EduGenie is a lightweight, AI-powered educational assistant built to simplify and enhance the learning experience through the power of Generative AI. Developed primarily for students, self-learners, and educators, the platform provides intelligent support by breaking down complex concepts, answering academic questions, generating customized quizzes, summarizing dense text, and creating personalized step-by-step learning roadmaps. The system architecture combines an asynchronous FastAPI backend with a clean HTML, CSS, and Jinja2 frontend, designed specifically to operate smoothly on resource-constrained hardware such as the Apple Mac M1. To balance efficiency with cognitive performance, EduGenie integrates a hybrid AI pipeline using local model weights from LaMini-Flan-T5 alongside cloud-based inference from the Google Gemini 1.5 Pro API. The codebase is organized within an structured development framework utilizing Git and GitHub for version control, while a relational database layer handles user session data, historical metrics, and performance analytics securely.

| S.No | Architecture Component / Layer | Technology Chosen | Justification / Purpose |
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| **1** | **Frontend / Client-Side**<br><br>*(e.g., React, Angular, HTML/CSS)* | HTML, CSS, Jinja2 Templates | Provides a lightweight, responsive interface that loads immediately without heavy JavaScript framework overhead. Jinja2 handles dynamic server-side data injection seamlessly. |
| **2** | **Backend / Server-Side**<br><br>*(e.g., Node.js, Python/Django, Java)* | FastAPI, Uvicorn | Selected for its asynchronous capabilities, fast execution speed, automatic OpenAPI documentation, and minimal boilerplate code when managing concurrent AI requests. |

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| **3** | **Database / Data Storage**<br><br>*(e.g., MySQL, MongoDB, PostgreSQL)* | SQLite/PostgreSQL | Chosen because it handles structured relational mapping effectively. It safely tracks user profiles, stores historically generated study roadmaps, and saves quiz performance scores with data integrity. |
| **4** | **Cloud / Hosting / Deployment**<br><br>*(e.g., AWS, Azure, Heroku, Docker)* | Local Machine(Mac M1 optimization/Docker) | Selected because the optimized framework allows cost-effective local execution on resource-constrained devices, while Docker readiness ensures simple, scalable deployment to cloud servers if needed later. |
| **5** | **Version Control & CI/CD** | Git & Github | Chosen to enable seamless codebase collaboration between team members, precise version history tracking, and structured deployment workflows. |
| | *(e.g., GitHub, GitLab, Jenkins)* | | |

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| **6** | **Third-Party APIs / Other Tools**<br><br>*(e.g., Stripe, Twilio, Google Maps)* | Google Generative AI SDK (Gemini API) | Enables core generative capabilities in the project, passing structured prompt payloads to deliver advanced question answering and roadmap layouts. |